

CODE-FILES

Overall System Kya Hai?

Ye project:

- **Frontend (React/Next.js)** → User interface (pages, buttons, forms, upload, dashboard)
 - **Backend (Python FastAPI)** → Image processing, preprocessing pipeline, API
 - **Goal** → MRI image upload → preprocessing → result/metadata return
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Frontend (my-app folder)

CSS / UI ka kaam

Tumhare project me:

- UI components bane hue hain:
 - Button.tsx
 - Card.tsx
 - Input.tsx
- Ye **reusable components** hain:
 - Matlab ek button bana ke har jaga use kar sakte ho
- Styling:
 - Modern UI ke liye Tailwind / CSS classes use hoti hain
 - Layout: Header, Sidebar, Dashboard waghera

Teacher ko bolo:

“Sir, humne component-based UI banayi hai jahan Button, Card, Input jaise reusable components use kiye hain taake code clean aur maintainable rahe.”

Components kya hotay hain?

Components = chhoti chhoti UI blocks

Example:

- Header.tsx → upar ka header
- Sidebar.tsx → side menu
- Button.tsx → har jaga use hone wala button

Faida:

- Code repeat nahi hota
 - Easy to manage
 - Professional React approach
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Pages / Flow

User:

1. Login / Dashboard pe jata
 2. MRI image upload karta
 3. Frontend image ko **backend API** ko bhejta
 4. Result show hota
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Models (Frontend) – models/ folder

Yahan:

- User.ts
- Patient.ts
- PatientHistory.ts

Ye **TypeScript models** hain:

Matlab:

- Ye define karte hain ke:
 - User ka data kaisa hoga
 - Patient ka record kaisa hoga

- History ka structure kya hoga

👉 **Teacher ko bolo:**

“Sir, frontend me humne TypeScript models banaye hain taake data ka structure fixed rahe aur errors kam hon.”

5 lib folder

- db.ts → database connection / helper
- reportGenerator.ts → report banane ka logic
- notifications.ts → messages / alerts

Matlab:

Ye **helper functions** hain jo app ke different parts me use hotay hain.

⚙️ **Backend (backend folder)**

Backend Python me bana hai using **FastAPI**.

Main file:

👉 main.py

Is ka kaam:

- API banana
 - Image receive karna
 - Preprocessing pipeline run karna
 - Result wapas frontend ko dena
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🌱 **Backend ka Flow (Simple Words)**

1. User image upload karta (frontend se)
2. Request backend ke main.py pe jati
3. Backend:
 - Image save karta

- Preprocessing pipeline chalata
 - 4. Pipeline:
 - Resize
 - Grayscale
 - Denoise
 - Skull strip
 - Histogram equalization
 - 5. Result + metadata frontend ko return
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Preprocessing Folder

pipeline.py

- Ye **poora sequence control** karta hai
- Matlab:
 - Step 1 call
 - Step 2 call
 - Step 3 call ...
- Ek **manager** jaisa kaam karta hai

steps.py

- Is me **actual functions** likhe hain:
 - `resize_image()`
 - `to_grayscale()`
 - `denoise()`
 - `skull_strip()`
 - `histogram_equalize()`

👉 Teacher ko bolo:

“Sir, preprocessing ko humne modular banaya hai: ek pipeline file hai jo steps ko control karti hai, aur ek steps file hai jahan actual image processing functions likhe hue hain.”

🧠 Backend Functions kya karte hain?

Har function:

- Image leta hai
- Us pe ek operation karta hai
- Updated image next step ko de deta hai

Example:

- `resize()` → image 256x256 karta
 - `grayscale()` → color hata ke intensity pe lata
 - `denoise()` → noise remove karta
 - `skull_strip()` → skull remove karta
 - `hist_eq()` → contrast improve karta
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🌐 API ka Concept

FastAPI me:

- Endpoint bana hota hai jaise:
/upload ya /process
- Frontend:
 - Image bhejta hai
- Backend:
 - Process karta
 - JSON response bhejta hai

Response me:

- Status

- Kaun kaun se steps apply hue
 - Result info
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Kya Backend me Database Model use hua?

Backend side pe:

- Abhi main focus **image preprocessing** pe hai
- Database zyada frontend / Next.js side pe handle ho raha (MongoDB + Mongoose)

Frontend me:

- User
- Patient
- PatientHistory

Ye **records store** karne ke liye hain.

Teacher ko bolo:

“Sir, patient aur user related models frontend/DB side pe hain, jabke backend ka main focus MRI preprocessing pipeline hai.”

__init__.py kis liye hoti hai?

◆ Simple words me:

__init__.py Python ko batati hai ke:

“Ye folder ek **package/module** hai.”

Matlab:

- Jab tum kisi folder ko import karte ho:
- from preprocessing import pipeline
- To Python tab samajhta hai ke **preprocessing** ek proper module hai, kyun?
 Kyun ke us folder ke andar __init__.py hoti hai.

◆ Tumhare project me:

- preprocessing/ folder ke andar `__init__.py` hai
- Is ka kaam:
 - Folder ko **importable module** banana
 - Code ko **clean structure** dena
 - Future me easily:
 - `from preprocessing.steps import resize_image`

jaisa import possible banana

 **Viva line:**

“Sir, `__init__.py` Python ko batati hai ke ye folder ek package hai aur is se hum preprocessing ke modules ko easily import kar sakte hain.”

 **steps.py** kis liye use hoti hai?

◆ **Simple words me:**

steps.py me **actual image processing ke functions** likhe hue hain.

Jaise:

- resize image
- grayscale conversion
- denoise
- skull stripping
- histogram equalization

Har function:

- Image leta hai
 - Ek **specific kaam** karta hai
 - Updated image return karta hai
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Testing ka Idea

- Black box testing
 - Matlab:
 - Input do (image upload)
 - Output dekho (processed result)
 - UI crash na ho
 - Correct response aaye
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Viva Ready Short Answers

Q: CSS / UI kaise banayi?

👉 Component-based UI using React/Next.js, reusable Button, Card, Input components.

Q: Backend kis me hai?

👉 Python FastAPI me.

Q: Backend ka main kaam?

👉 Image receive karna, preprocessing pipeline chalana, result return karna.

Q: Pipeline kya hai?

👉 Steps ka sequence jo image pe processing karta hai (resize, grayscale, denoise, etc.)

Q: Models use hue?

👉 Frontend/DB side pe User, Patient, PatientHistory models use hue hain.

Q: API ka role?

👉 Frontend aur backend ke darmiyan communication.

One-Line Summary (Teacher ke liye)

“Sir, hamara system React/Next.js based frontend hai jahan reusable UI components aur TypeScript models hain, aur Python FastAPI based backend hai jo MRI images ko ek modular preprocessing pipeline ke through process karta hai aur API ke zariye results frontend ko return karta hai.”

LIBRARIES

Frontend Libraries (React / Next.js side)

✓ 1. React

- UI banane ke liye
- Components (Button, Card, Pages, etc.) banane ke liye
- Single Page Application ka structure

👉 Viva line:

“Sir, React frontend UI aur component-based structure ke liye use hua hai.”

✓ 2. Next.js

- React ka framework
- Routing (pages banana)
- Better performance aur structure

👉 Viva line:

“Sir, Next.js React ke upar use hua hai for routing aur better project structure.”

✓ 3. TypeScript

- JavaScript ka strict version
- Models banane ke liye (User, Patient, etc.)
- Errors kam karne ke liye

👉 Viva line:

“Sir, TypeScript type safety aur models define karne ke liye use kiya hai.”

✓ 4. Tailwind CSS / CSS

- UI styling ke liye
- Buttons, cards, layout, colors, spacing

👉 **Viva line:**

“Sir, UI styling ke liye Tailwind/CSS use kiya hai taake responsive aur clean design mile.”

✅ 5. Axios / Fetch (API calls)

- Frontend se backend ko request bhejne ke liye
- Image upload aur result receive karne ke liye

👉 **Viva line:**

“Sir, frontend se backend API call karne ke liye Axios/Fetch use kiya hai.”

⚙️ Backend Libraries (Python / FastAPI side)

✅ 1. FastAPI

- Backend API banane ke liye
- Endpoints jaise /upload, /process
- Frontend se requests receive karne ke liye

👉 **Viva line:**

“Sir, FastAPI backend framework hai jo API endpoints aur request handling ke liye use hua hai.”

✅ 2. Uvicorn

- FastAPI server run karne ke liye

👉 **Viva line:**

“Sir, Uvicorn FastAPI server run karne ke liye use hota hai.”

✅ 3. NumPy

- Image data ko array form me handle karne ke liye
- Fast numerical operations ke liye

👉 **Viva line:**

“Sir, NumPy image data aur numerical processing ke liye use hui hai.”

✅ **4. OpenCV (cv2)**

- Image processing ke liye:
 - Resize
 - Grayscale
 - Denoise
 - Basic image operations

👉 **Viva line:**

“Sir, OpenCV image preprocessing operations jaise resizing aur denoising ke liye use hui hai.”

✅ **5. Pillow (PIL)**

- Image load aur save karne ke liye
- Format handling (JPG, PNG, etc.)

👉 **Viva line:**

“Sir, Pillow image read/write aur format handling ke liye use hui hai.”

✅ **6. scikit-image (skimage) (agar use ho rahi ho)**

- Advanced image processing:
 - Histogram equalization
 - Filtering
 - Enhancement

👉 **Viva line:**

“Sir, scikit-image advanced image enhancement operations ke liye use hui hai.”

✅ **7. pydantic**

- FastAPI me data validation ke liye
- Request / response models ke liye

👉 **Viva line:**

“Sir, Pydantic data validation aur schema define karne ke liye use hoti hai.”

Database Side

✅ 8. MongoDB + Mongoose (Frontend/Node side)

- User, Patient, History store karne ke liye
- Models:
 - User
 - Patient
 - PatientHistory

👉 **Viva line:**

“Sir, MongoDB database hai aur Mongoose ORM hai jo user aur patient records store karne ke liye use hua hai.”